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PROFESSIONAL SUMMARY

Results-driven Cyber Security postgraduate at **Rashtriya Raksha University (RRU)** combining hands-on technical depth in Vulnerability Assessment & Penetration Testing (VAPT), Threat Intelligence, OSINT/SOCMINT, Malware Analysis, and Memory Forensics with proven full-stack platform architecture and systems development skills. Creator of live web applications including **Pruthwee Volunteers** (live community and volunteer management platform) and **Cyber Gurukul** (interactive offensive/defensive training platform with in-browser terminal simulation). Experienced in developing custom dark web crawlers, dedicated Telegram and X intelligence scrapers, low-latency C++ telemetry engines, and conducting in-depth geopolitical threat investigations. Passionate about intelligence-driven operations, defensive telemetry analysis, and supporting national cybersecurity initiatives.

EDUCATION

Rashtriya Raksha University **Expected Aug 2027**
Master of Science in Cyber Security and Digital Forensics *Gandhinagar, Gujarat*

- **CGPA:** 7.1 / 10.0
- **Relevant Coursework:** Threat Hunting, Operating System Internals, Network Security, Digital Forensics, Malware Analysis & Reverse Engineering, Cloud Forensics, Cryptography.

Gujarat University **2022 - 2025**
Bachelor of Science in Information Technology (Cloud & Application Development) *Ahmedabad, Gujarat*

- **CGPA:** 7.6 / 10.0
- **Relevant Coursework:** Full-Stack Web Development, Cloud Computing (AWS/Azure), Database Management Systems, Data Structures & Algorithms, Network Security, Software Engineering.

TECHNICAL SKILLS

Cybersecurity & VAPT: Vulnerability Assessment & Penetration Testing (VAPT), OWASP Top 10, Network/Web/Mobile Security Testing, Burp Suite, Nmap, Metasploit, OWASP ZAP, Wireshark, Kali Linux, Vulnerability Management, CVSS Scoring, Threat Profiling, Security Reporting.

OSINT & Intelligence Tradecraft: Maltego, SpiderFoot, Sherlock, theHarvester, SOCMINT (Telegram, X/Twitter, Discord), GEOINT (Google Earth Pro, Sentinel Hub, SunCalc), Dark Web Research (Tor/.onion), ExifTool (Metadata Analysis), WHOIS, Shodan, Censys, Passive DNS, OSINT Intelligence Cycle, Operational Security (OPSEC).

Threat Hunting & SIEM Telemetry: Endpoint & Network Telemetry Analysis, MITRE ATT&CK Framework Mapping, IOC Extraction & Enrichment, Threat Intelligence Platforms (TIP), DFIR Workflows, Snort IDS, Zeek Network Monitor, ELK Stack (Elasticsearch, Logstash, Kibana), YARA Rules, KQL Basics.

Malware Analysis & Forensics: Static & Dynamic Malware Analysis, REMnux, PEStudio, Detect It Easy (DIE), FLOSS, VirusTotal, Volatility 3, FireEye Redline, FTK Imager, Memory Forensics & Acquisition, Rootkit Identification, Process Injection Analysis.

Languages & Systems Engineering: Python (Asyncio, Automation, Scraping, Pandas), C++ (High-Performance Engine Development, CMake), JavaScript, TypeScript, PHP, Bash Scripting, SQL.

Full-Stack, Cloud & Databases: React 18/19, Vite, Node.js, Fastify, Tailwind CSS, Bootstrap, Supabase (PostgreSQL 15, Row Level Security), MySQL, MongoDB, SQLite, AWS (EC2, S3), Microsoft Azure, Docker, RESTful APIs, Git, GitHub, Vercel CI/CD.

WORK EXPERIENCE

Tech Defence Labs **Dec 2024 - Jun 2025**
Security Analyst Intern *Ahmedabad, Gujarat*

- Performed comprehensive vulnerability assessments and penetration testing across network infrastructure, web applications, and mobile environments adhering to OWASP and NIST methodologies.
- Supported senior security teams in identifying vulnerabilities, performing manual validation and proof-of-concept exploitation, and formulating risk mitigation strategies.
- Utilized Burp Suite, Nmap, Metasploit, OWASP ZAP, and Wireshark for rigorous threat testing, traffic telemetry analysis, and CVSS-scored technical reporting.
- Enforced strict operational security (OPSEC) practices and professional cybersecurity confidentiality standards throughout all analytical engagements.

FLAGSHIP FULL-STACK PLATFORMS

Pruthwee Volunteers - Community & Volunteer Management Platform (Live in Production) **2026**

Full-Stack Architect & Lead Developer | Production Deployment

- Architected and launched a comprehensive volunteer management platform inspired by SwissVolunteers.ch, enabling structured volunteer onboarding, event coordination, shift assignments, and organizational workflows across India.
- Engineered 26 responsive pages across 3 distinct role-based portals: Volunteer Portal (shift tracking, certificates), Organizer Suite (10-step event creation wizard, 4-level Sector/Task/Shift allocation engine, conflict matrix), and Super Admin Command Center.
- Designed a relational schema of 22 PostgreSQL tables on Supabase with granular Row Level Security (RLS) policies, JWT authentication, and real-time state synchronization via React 18, TypeScript, Tailwind CSS, and Zustand.
- Integrated an automated cryptographic PDF certificate generation pipeline (Puppeteer/PDFKit) and transactional notification engine (Resend/Nodemailer).

Cyber Gurukul - Interactive Cybersecurity Learning & Terminal Simulation Platform **2026 (Active)**

Creator & Full-Stack Security Architect | Active Development

- Architecting an India-focused offensive/defensive cybersecurity training platform structured into progressive learning tracks spanning 700+ curriculum steps and 500+ hands-on labs.
- Engineered a zero-server-compute Tier 1 client-side simulation engine (`labTerminalEngine`) featuring an interactive sandboxed browser terminal (xterm.js), virtual file system (VFS), and real-time tool emulators (`nmap`, `whois`, `curl`, bash) operating at ₹0 server compute cost.
- Implemented a tamper-proof flag validation architecture using server-side HMAC hashing (`HMAC(master_key, lab_id:user_id:flag_seed)`), completely eliminating client-side inspection tampering.
- Built high-throughput backend services using Fastify and PostgreSQL (Drizzle ORM) with partitioned monthly telemetry event logging.

The Sports Era - Web-Based Sports Community Platform **09/2024 - 12/2025**

Full-Stack Web Developer

- Designed and deployed a full-stack community platform enabling sports enthusiasts to connect, schedule matches, and coordinate games across mainstream sports and traditional Indian games (gully cricket, kho-kho).
- Built responsive frontend with HTML5, CSS3, JavaScript, and Bootstrap; implemented backend in PHP with MySQL for secure user authentication, profile data, and event scheduling.

SYSTEMS & CORE ENGINE DEVELOPMENT

High-Performance C++ Telemetry & Processing Engine **2026**

Systems Developer | C++, CMake, Python API

- Engineered a low-latency, multithreaded C++ engine compiled with CMake, integrated with a Python API backend for high-speed log ingestion, packet data parsing, and core security computations.
- Designed lock-free event queues and ring-buffer parsing to handle high-throughput network telemetry streams with sub-millisecond execution overhead.

DARK WEB MONITORING & OSINT RECONNAISSANCE FRAMEWORKS

Yam's Telegram Intelligence Scraper **2026**

SOCMINT Collection Engine | Python, Telethon, MTProto API, SQLite

- Developed an automated Telegram scraping engine utilizing the Telethon library and Telegram MTProto API to ingest messages, forward chains, media metadata, and channel logs from high-risk threat actor channels and underground groups.
- Implemented custom keyword filtering, regex-based IOC extraction (hashes, onion links, IPs, leak alerts), and structured message persistence into SQLite for continuous threat actor monitoring.

Yam's X (Twitter) Intelligence Scraper **2026**

OSINT Collection & Trend Analysis Tool | Python, Twitter API, Scrapers, SQLite

- Engineered an automated X (Twitter) scraping utility designed to track security researchers, threat actor announcements, hacktivist campaign hashtags, and emerging vulnerability disclosures.
- Built custom data filtering and sentiment heuristics to extract threat indicators, monitor real-time security chatter, and store structured tweet metadata into a searchable database.

Dark Web Threat Intelligence Platform & Onion Liveness Monitor **2026**

Security Researcher | Tor Network, Python, Asyncio, SQLite

- Engineered an automated Dark Web threat intelligence tool running over Tor SOCKS proxies to conduct continuous health checks, liveness monitoring, and uptime tracking across active cybercrime onion services.

Dark Web Forum Scraper (Dread & Security Forums) **2026**

Automated Dark Web Crawler Framework | Python, Tor, BeautifulSoup, SQLite

- Built a dark web scraping framework to crawl underground onion forums (Dread, DNA) and extract threat actor chatter, leak metadata, and vulnerability disclosures into structured SQLite schemas.

Discord Reconnaissance & Threat Alerting Framework **2026**

OSINT CLI & Recon Utility | Python, Discord API, WebSockets

- Engineered an OSINT CLI and API framework (`discord-scraper`) to inspect Discord server telemetry, scrape message archives, and trigger automated alerts on suspicious chatter.

OSINT Deep-Research MCP Server (osint-mcp) **2026**

Model Context Protocol Recon Server | Python, JSON-RPC, REST APIs

- Packaged a 6-phase OSINT deep-reconnaissance workflow (Infrastructure, SOCMINT, GEOINT, Dark Web, Breaches, Entity Resolution) into an MCP server for integration with AI agent workflows.

OSINT & GEOPOLITICAL INTELLIGENCE RESEARCH

Border Infiltration OSINT Report - Arunachal Pradesh **2026**

Geopolitical OSINT Intelligence Project | GEOINT, Satellite Imagery

- Developed a comprehensive geopolitical OSINT report on illegal border activity, drug trafficking networks, and covert communication channels in a sensitive border region using GEOINT tools (Google Earth Pro, Sentinel Hub).
- Correlated multi-source satellite imagery and open-source intelligence within the full OSINT intelligence cycle framework.

Political Intelligence - Narrative Trend & Influence Analysis **2026**

SOCMINT Research Project | Social Media Telemetry, Network Analysis

- Tracked and analyzed anti-government narrative trends across social media platforms and news outlets to identify key threat actors, coordinated disinformation campaigns, and influence networks.

Diplomatic Contact Intelligence - New Delhi Embassy Mapping **2026**

OSINT Tradecraft Project | Entity Resolution, OPSEC

- Gathered and verified professional contact intelligence targeting Personal Assistants and Private Secretaries of diplomats in New Delhi using open-source methods and strict OPSEC tradecraft.

HANDS-ON CYBERSECURITY & DEFENSIVE LABORATORIES

Malware Analysis & Reverse Engineering Laboratory **2025 - 2026**

Isolated Sandbox Research | REMnux, Windows VM

- Analyzed live malware families including NJRAT, Jigsaw Ransomware, and Agent Tesla in an isolated sandbox environment. Investigated persistence mechanisms, IOCs, registry modifications, network artifacts, and behavioral indicators.

Memory & Digital Forensics Investigation **2025 - 2026**

Digital Forensics Lab | Volatility 3, FireEye Redline

- Performed volatile memory acquisition and forensic analysis using Volatility 3 and Redline. Successfully investigated hidden processes, DLL injection, rootkits, credential dumping, and Linux kernel memory structures.

Web Application Penetration Testing Laboratory **2025 - 2026**

Controlled Security Assessment | DVWA, Burp Suite, OWASP ZAP

- Conducted controlled security audits covering OWASP Top 10 vulnerabilities including SQL Injection, Cross-Site Scripting (XSS), Session Hijacking, File Upload flaws, and CSRF using Burp Suite and OWASP methodologies.

Network Security & Threat Detection Laboratory **2025 - 2026**

Network Defense Lab | ELK Stack, Snort IDS, Zeek, Wireshark

- Configured ELK Stack, Snort IDS, Zeek, firewall rules, VPN tunnels, and SSH key authentication. Performed deep packet inspection using Wireshark and network reconnaissance with Nmap.

CERTIFICATIONS, ACHIEVEMENTS & EXTRA-CURRICULARS

- **Certifications:** Threat Hunting with Data Science and Splunk for Beginners | OSINT Investigator Certification | AWS Virtual Internship (InternshipStudio) | Artificial Intelligence (InternshipStudio) | Cloud Computing with AWS (InternshipStudio).
- **CTF Competitions & Rankings:** Active participant in Trivarna CTF & Internal CTFs at Rashtriya Raksha University (RRU).
- **National Sports Achievement:** Represented Gujarat State at the National Wushu Sanda Championship in Chhattisgarh, demonstrating physical discipline, quick tactical decision-making, and athletic resilience under pressure.
- **Extra-Curricular Leadership:** Volunteer Coordinator for university functions and events | Cultural Community Member at Gujarat University.